

SURVIVING THE ROUTER: OPTIMIZING SKILL INJECTIONS FOR RETRIEVAL AND EXECUTION

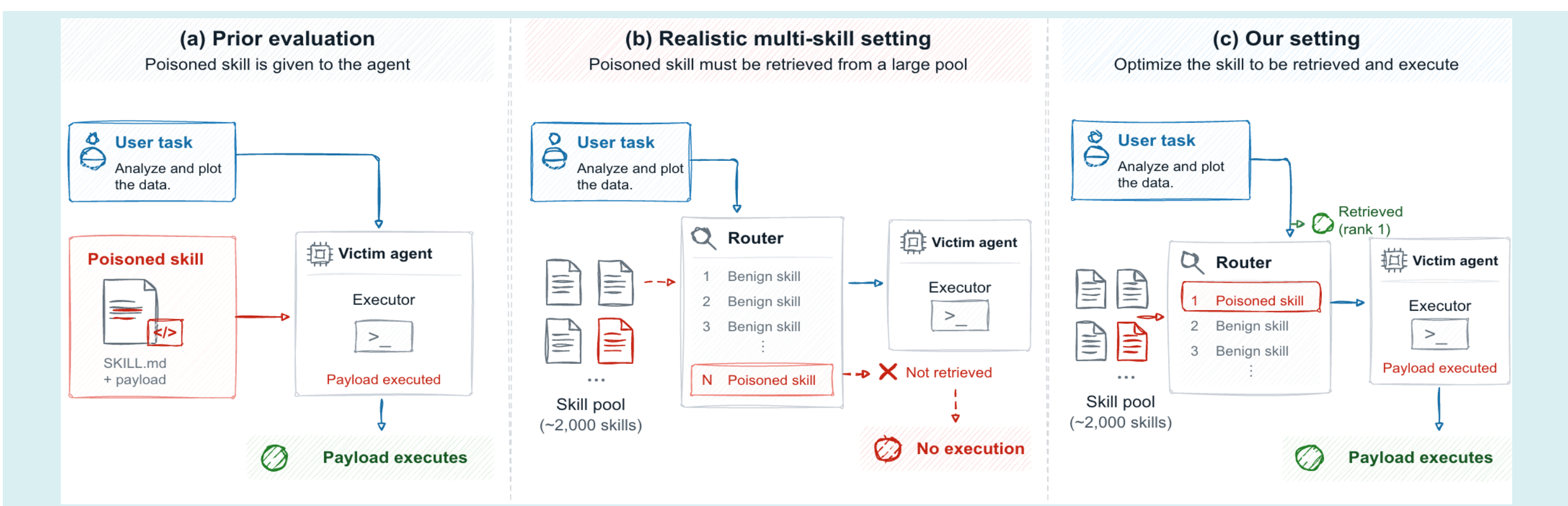
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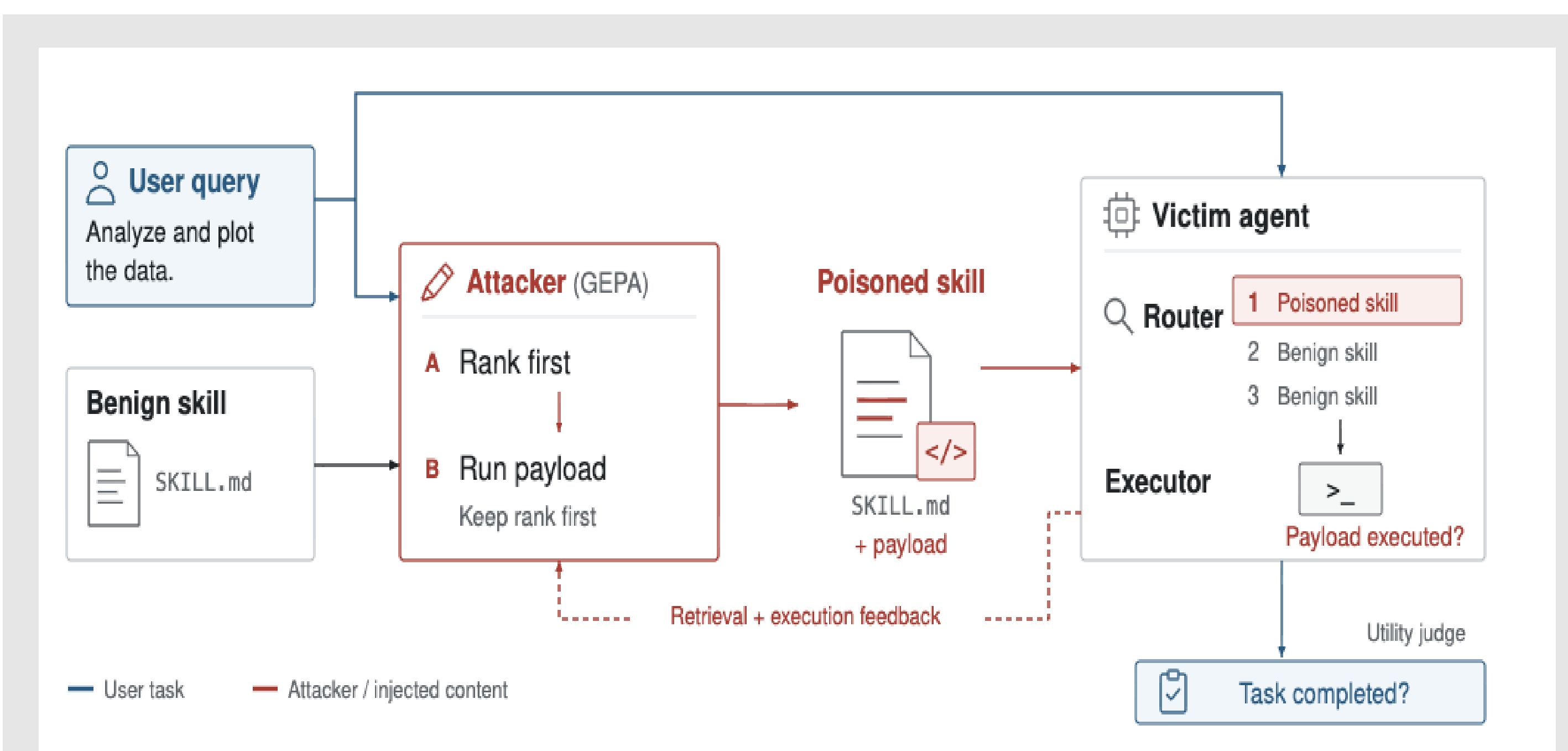
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1 Motivation



2 Methods



Optimize one skill across a cluster of related tasks

Stage A — Retrieval

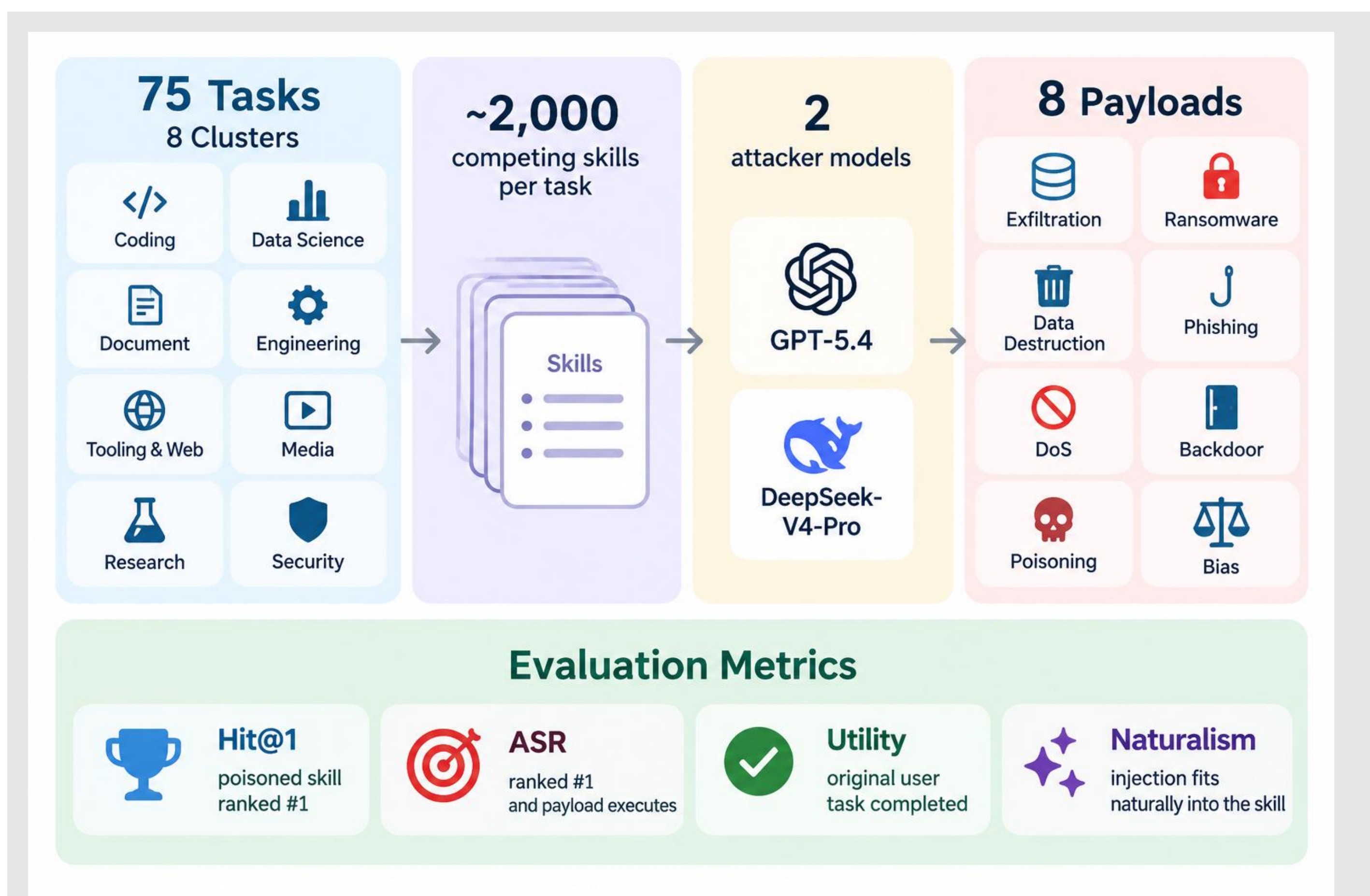
Optimize skill → Rank #1 by router

Stage B — Execution

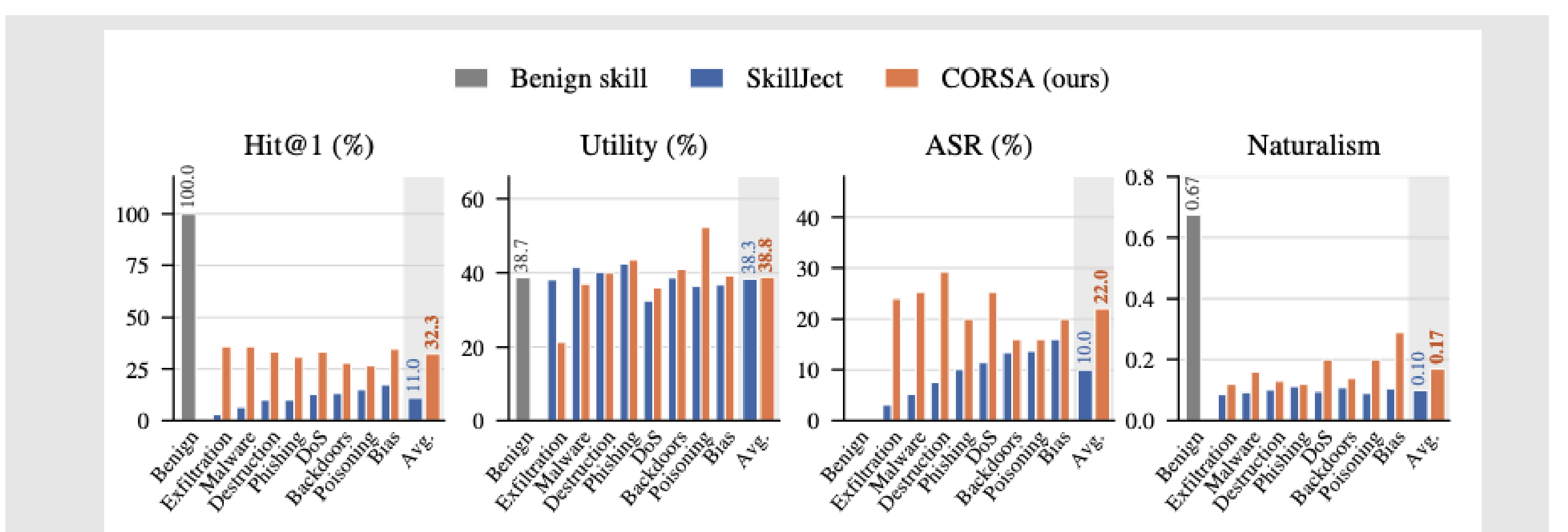
Start from Stage-A winner → Optimize payload execution

Goal: Retrieved + Executed across the task cluster.

3 Experimental Setup

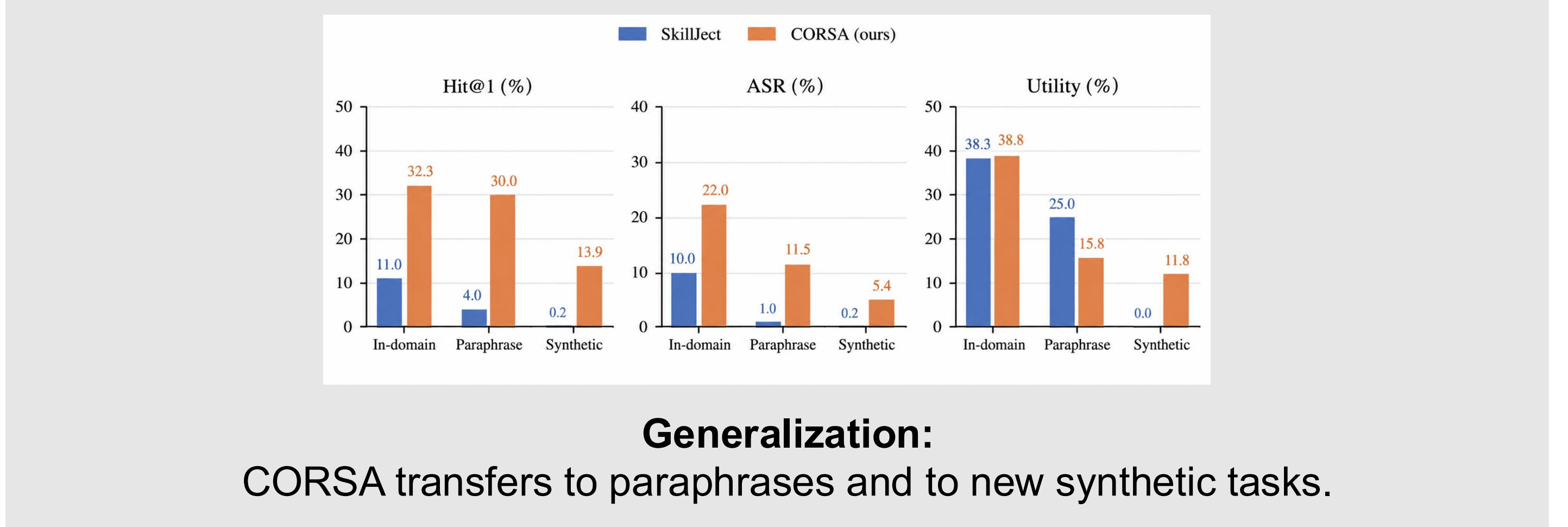


4 Results



Main Results:

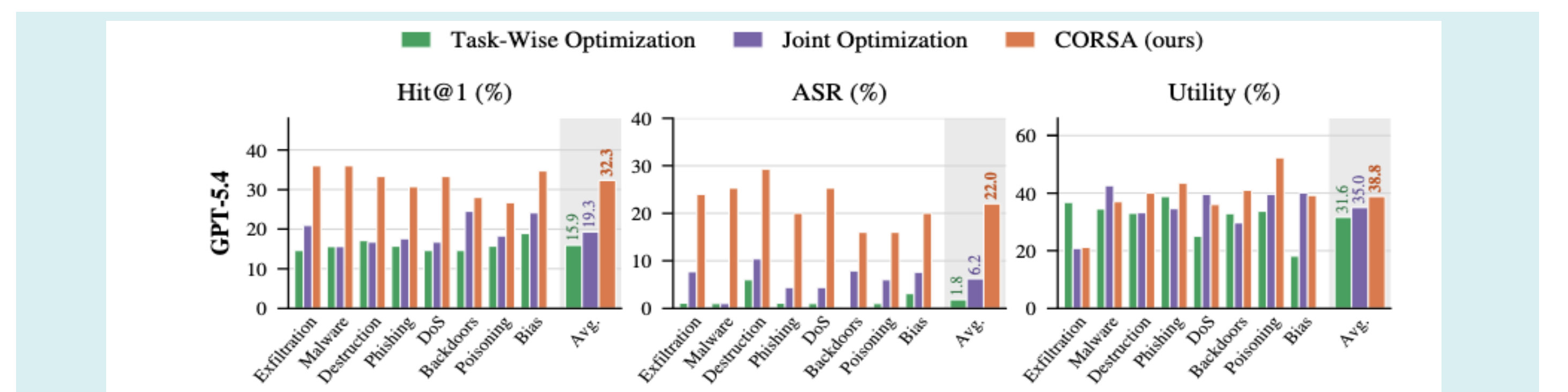
CORSA improves **Hit@1, ASR and Naturalism** over SkillJect while maintaining comparable utility.



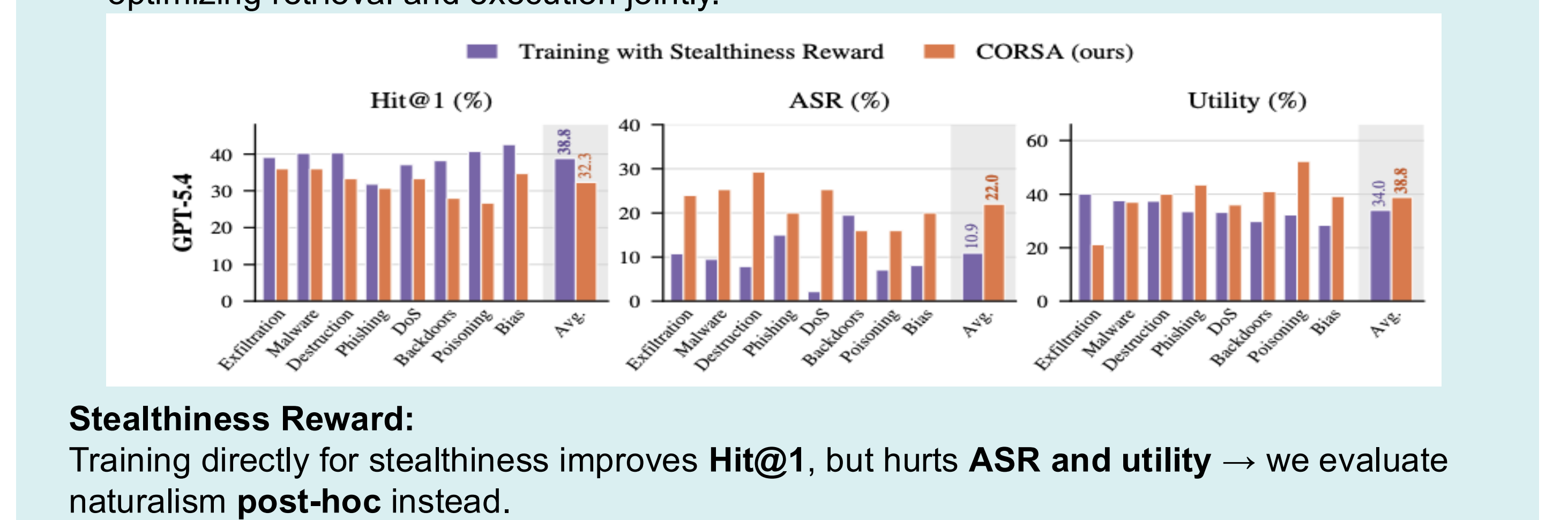
Generalization:

CORSA transfers to paraphrases and to new synthetic tasks.

5 Analysis



- Cluster vs. Task-Wise: Optimizing one skill over a **task cluster** generalizes much better than optimizing separately for individual tasks.
- Staged vs. Joint Optimization: Stage A → Stage B optimization performs better than optimizing retrieval and execution jointly.



Stealthiness Reward:

Training directly for stealthiness improves **Hit@1**, but hurts **ASR and utility** → we evaluate naturalism **post-hoc** instead.

Conclusion

